

ECEN 662:

Estimation & Detection Theory

Lecture 5: Uniformly Minimum-Variance Unbiased Estimation (Part II)

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UMVUE without a complete sufficient statistic

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- While in our previous study, the model parameter space is $\Theta = (0, \infty)$. In this study, we shall assume that $\Theta = (1, \infty)$, i.e., we have the additional knowledge that the unknown parameter $\theta > 1$
- What difference does this additional knowledge make?

- Recall that under the previous assumption that $\Theta = (0, \infty)$,

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- As a result, the unbiased estimator

$$\hat{\theta} = \frac{n+1}{n} T$$

is an UMVUE of θ with the minimum variance given by

$$\text{Var}_{\theta}[\hat{\theta}] = \frac{\theta^2}{n(n+2)}$$

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- For any $\theta > 1$, the expected value of $\hat{\theta}'$ can be calculated as

$$\begin{aligned} \mathbb{E}_{\theta}[\hat{\theta}'] &= \int_0^1 f_{\theta}(t)dt + \int_1^{\theta} \frac{n+1}{n}t f_{\theta}(t)dt \\ &= n\theta^{-n} \int_0^1 t^{n-1}dt + (n+1)\theta^{-n} \int_1^{\theta} t^n dt \\ &= \theta^{-n} + \theta^{-n}(\theta^{n+1} - 1) \\ &= \theta \end{aligned}$$

- That is,

$$\hat{\theta}' = \begin{cases} 1, & \text{if } T \leq 1 \\ \frac{n+1}{n}T, & \text{if } T > 1 \end{cases}$$

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- Let $\phi(T)$ be a function of T such that

$$\mathbb{E}_{\theta}[\phi(T)] = 0, \quad \forall \theta > 1$$

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- Let $\phi(T)$ be a function of T such that

$$\mathbb{E}_{\theta}[\phi(T)] = 0, \quad \forall \theta > 1$$

- As before, it can be shown that we *must* have

$$\phi(t) = 0, \quad \forall t > 1$$

- However, for $t \leq 1$, $\phi(t)$ does *not* have to be a zero function to satisfy

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- For example, if we let

$$\phi(t) = \begin{cases} t - \frac{n}{n+1}, & \text{if } t \leq 1 \\ 0, & \text{if } t > 1, \end{cases}$$

we have for any $\theta > 1$,

$$\begin{aligned} \mathbb{E}_\theta[\phi(T)] &= \int_0^1 \left(t - \frac{n}{n+1} \right) f_\theta(t) dt \\ &= n\theta^{-n} \int_0^1 t^{n-1} dt - (n+1)\theta^{-n} \int_0^1 t^n dt \\ &= \theta^{-n} - \theta^{-n} \\ &= 0 \end{aligned}$$

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- To make matters worse, it can be shown that T remains to be a *minimal* sufficient statistic, so in this case a complete sufficient statistic does *not* exist
- Is the (very sensible) $\hat{\theta}'$ an UMVUE in this case?
- Next, we shall provide an alternative characterization of UMVUE based on the notion of *unbiased estimators of 0* and use this characterization to show that $\hat{\theta}'$ is indeed an UMVUE of θ , knowing that $\theta > 1$

Variance reduction via unbiased estimators of 0

- Mathematically, for a given likelihood family $f_{\theta}(x)$, a statistic $U = U(X)$ is called an *unbiased estimator of 0* if $\mathbb{E}_{\theta}[U] = 0$ for *all* $\theta \in \Theta$

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- Let $\delta_0 = \delta_0(X)$ be an unbiased estimator of $g(\theta)$ and let $U = U(X)$ be an unbiased estimator of 0
- Note that for any real α ,

$$\delta_\alpha = \delta_0 + \alpha U$$

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- Further note that

$$\begin{aligned}\mathrm{Var}_\theta[\delta_\alpha] &= \mathrm{Var}_\theta[\delta_0 + \alpha U] \\ &= \mathrm{Var}_\theta[\delta_0] + \alpha^2 \mathrm{Var}_\theta[U] + 2\alpha \mathrm{Cov}_\theta[\delta_0, U],\end{aligned}$$

which is a *quadratic* function of α

- Let

$$g(\alpha) := \text{Var}_\theta[\delta_\alpha] = \text{Var}_\theta[\delta_0] + \alpha^2 \text{Var}_\theta[U] + 2\alpha \text{Cov}_\theta[\delta_0, U]$$

and note that

$$g(0) = \text{Var}_\theta[\delta_0] \quad \text{and} \quad g'(0) = 2\text{Cov}_\theta[\delta_0, U]$$

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we can always find a nonzero α such that

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- Therefore, for any unbiased estimator δ_0 of $g(\theta)$, if there exists an unbiased estimator U of 0 such that

$$\text{Cov}_\theta[\delta_0, U] \neq 0,$$

then δ_0 is *not* an UMVUE of $g(\theta)$

UMVUE and unbiased estimators of 0

- That is,

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- Let δ be any unbiased estimator $g(\theta)$. Then $\delta - \delta_0$ is an unbiased estimator of 0 and hence by our assumption,

$$\text{Cov}_\theta[\delta_0, \delta - \delta_0] = 0, \quad \forall \theta \in \Theta$$

- It follows that

$$\begin{aligned}\text{Var}_\theta[\delta] &= \text{Var}_\theta[(\delta_0 + (\delta - \delta_0))] \\ &= \text{Var}_\theta[\delta_0] + \text{Var}_\theta[\delta - \delta_0] + 2\text{Cov}_\theta[\delta_0, \delta - \delta_0] \\ &= \text{Var}_\theta[\delta_0] + \text{Var}_\theta[\delta - \delta_0] \\ &\geq \text{Var}_\theta[\delta_0]\end{aligned}$$

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- In light of the *Rao-Blackwell* theorem, to find an UMVUE, it suffices to consider estimators that are functions of a *sufficient* statistic. We have the following corollary to the previous theorem

- (Corollary) Let T be a sufficient statistic, and let $\delta_0 = \delta_0(T)$ be an unbiased estimator of $g(\theta)$. Then δ_0 is an UMVUE of $g(\theta)$ *if and only if*

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- Therefore, the above corollary immediately implies that any unbiased estimator that is a function of a *complete* sufficient statistic is an UMVUE
- (Proof of the corollary) The “only if” part of the corollary follows directly from the “only if” part of the theorem

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- We thus have

$$\begin{aligned}\text{Cov}_\theta[\delta_0, U'] &= \mathbb{E}_\theta[(\delta_0(T) - g(\theta))U'] \\ &= \mathbb{E}_\theta [\mathbb{E}_\theta[(\delta_0(T) - g(\theta))U'|T]] \\ &= \mathbb{E}_\theta [(\delta_0(T) - g(\theta))U''] \\ &= \text{Cov}_\theta[\delta_0, U''] \\ &= 0\end{aligned}$$

by the assumption that δ_0 is uncorrelated with any unbiased estimator of 0 that is also a function of T

- The fact that δ_0 is uncorrelated with any unbiased estimator of 0 (which may or may not be a function of T) proves that it is an UMVUE

Example: Uniform with unknown support ($\theta > 1$)

- Recall that for any $\theta > 1$, we have shown that

$$\hat{\theta}' = \begin{cases} 1, & \text{if } T \leq 1 \\ \frac{n+1}{n}T, & \text{if } T > 1 \end{cases}$$

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- To show that is is an UMVUE, it suffices to show that it is uncorrelated with any unbiased estimator of 0 that is a function of T
- Let $U = U(T)$ be an unbiased estimator of 0. For any $\theta > 1$, we have

$$0 = \mathbb{E}_{\theta}[U(T)] = \int_0^{\theta} U(t)f_{\theta}(t)dt = n\theta^{-n} \int_0^{\theta} U(t)t^{n-1}dt$$

and hence

$$\int_0^{\theta} U(t)t^{n-1}dt = 0$$

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- Combining the facts that $U(t) = 0$ for any $t > 1$ and $\int_0^\theta U(t)t^{n-1}dt = 0$ for any $\theta > 1$ gives

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- That is, for *any* unbiased estimator U of 0 that is a function of T , we must have 1) $U(t) = 0$ for any $t > 1$; and 2)

$$\int_0^1 U(t)t^{n-1}dt = 0$$

- For any $\theta > 1$, the covariance

$$\begin{aligned}\text{Cov}_\theta[\hat{\theta}', U] &= \mathbb{E}_\theta[(\hat{\theta}' - \theta)U] \\ &= \mathbb{E}_\theta[\hat{\theta}' U] \\ &= \int_0^\theta \hat{\theta}'(t) U(t) f_\theta(t) dt \\ &= \int_0^1 \hat{\theta}'(t) U(t) f_\theta(t) dt \\ &= n\theta^{-n} \int_0^1 U(t) t^{n-1} dt \\ &= 0\end{aligned}$$

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 &= 0
 \end{aligned}$$

- This completes the proof that

$$\hat{\theta}' = \begin{cases} 1, & \text{if } T \leq 1 \\ \frac{n+1}{n}T, & \text{if } T > 1 \end{cases}$$

is indeed an UMVUE of θ , knowing that $\theta > 1$

The uniqueness of UMVUE

- We conclude this lecture by proving that UMVUE is *unique* in the sense that if both $\delta(X)$ and $\delta'(X)$ are an UMVUE of $g(\theta)$, then we have

$$\mathbb{P}_\theta(\delta(X) = \delta'(X)) = 1, \quad \forall \theta \in \Theta$$

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- (Proof) Let $\phi = \delta(X) - \delta'(X)$. Since both $\delta(X)$ and $\delta'(X)$ are unbiased estimators of $g(\theta)$, ϕ is an unbiased estimator of 0

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- Since both $\delta(X)$ and $\delta'(X)$ are an UMVUE of $g(\theta)$, by the previous theorem, we must have

$$\text{Cov}_\theta[\delta(X), \phi] = 0 \quad \text{and} \quad \text{Cov}_\theta[\delta'(X), \phi] = 0$$

for all $\theta \in \Theta$

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for all $\theta \in \Theta$

- It follows that

$$\begin{aligned} \text{Var}_\theta[\phi] &= \mathbb{E}_\theta[\phi^2] = \mathbb{E}_\theta[(\delta(X) - \delta'(X))\phi] \\ &= \mathbb{E}_\theta[((\delta(X) - g(\theta)) - (\delta'(X) - g(\theta)))\phi] \\ &= \text{Cov}_\theta[\delta(X), \phi] - \text{Cov}_\theta[\delta'(X), \phi] = 0, \quad \forall \theta \in \Theta \end{aligned}$$

- We thus conclude that

$$\mathbb{P}_\theta[\phi = 0] = \mathbb{P}_\theta[\delta(X) = \delta'(X)] = 1$$

for all $\theta \in \Theta$